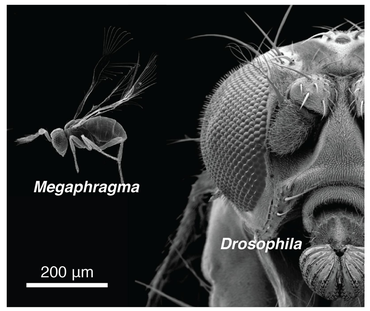
Sensory pathways to the *Megaphragma* central complex

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Introduction

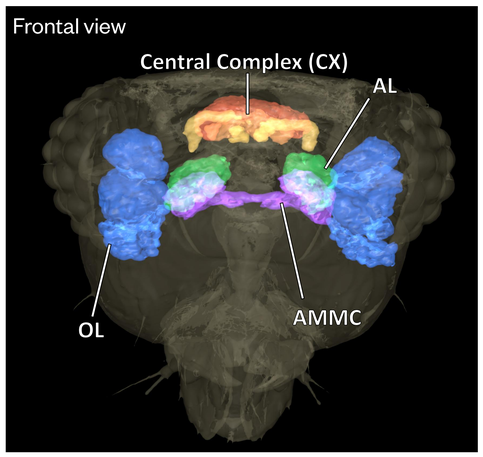
Megaphragma viggianii is a microscopic wasp whose brain contains only ~**9,400** neurons, about an order of magnitude smaller than that of *Drosophila.* Despite this extreme miniaturization, it performs complex behaviors such as flight and navigation, which are guided by the **central complex (CX)** in other insects [1].



We sought to answer:

How do different sensory modalities contribute to **central complex-dependent** goal-directed navigation in *Megaphragma*, as revealed by its connectome?

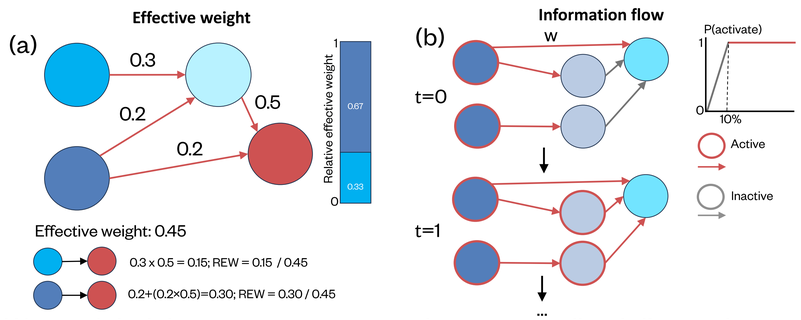
Methods



We started with three sensory input populations:

- Photoreceptors (**PR**) → optic lobe (**OL**)
- Olfactory receptor neurons (**ORN**) → antennal lobe (**AL**)
- Antennal mechanosensory neurons (**AN_mech**) → antennal mechanosensory and motor center (**AMMC**)

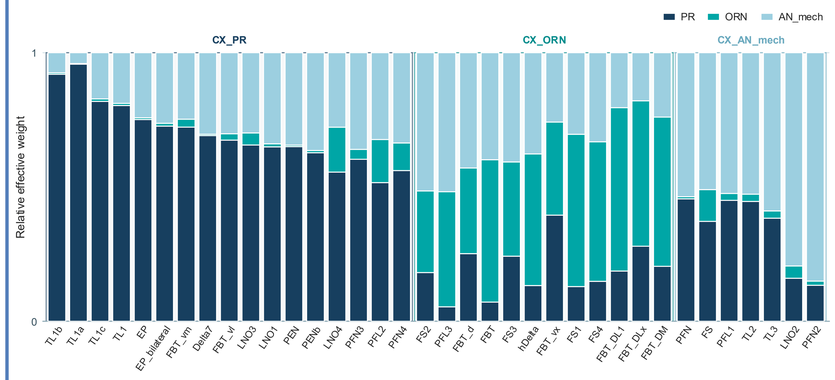
We traced signal propagation through the connectome using effective weight (a) [2] and information flow (b) [3], revealing downstream clusters and sensory pathways to the CX.



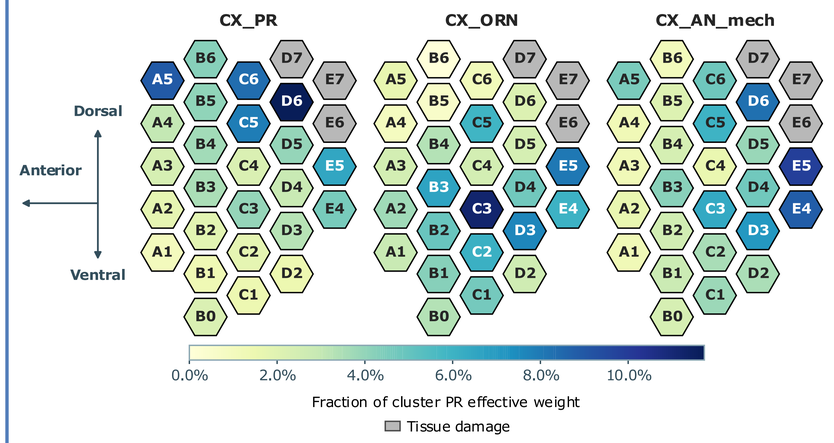
We modeled the connectome as an input-normalized directed adjacency matrix and analyzed it with [4].

Findings

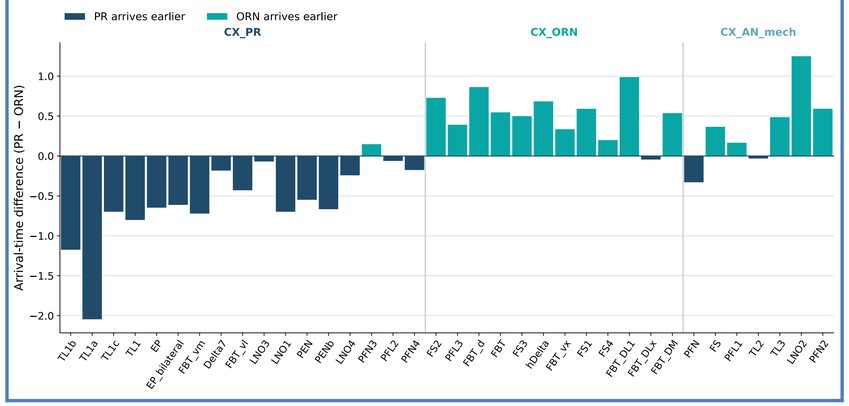
Relative effective weights reveal three clusters: **CX_PR**, **CX_ORN**, and **CX_AN_mech**, each dominated by a distinct sensory modality.



Normalized retinal input differs across the clusters: PR influence is weighted towards **dorsal** ommatidia in CX_PR, **ventral** ommatidia in CX_ORN, and intermediate in CX_AN_mech.

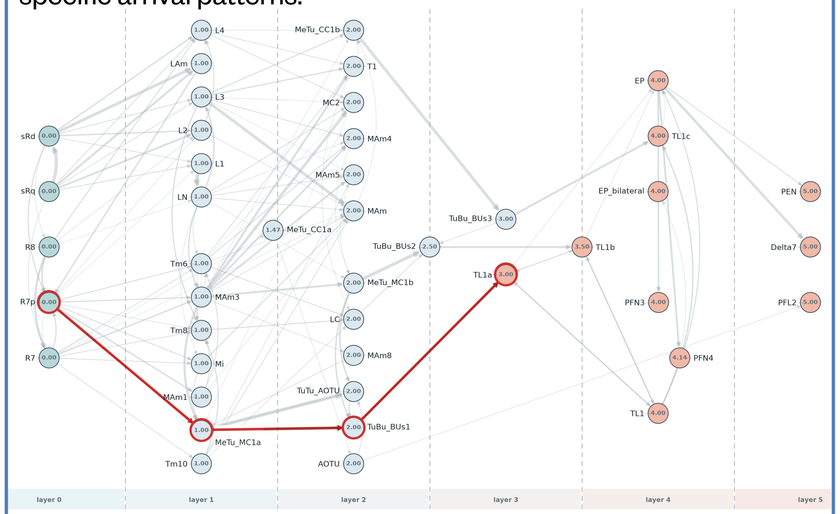


Mean modeled arrival differed across CX populations: PR input required fewer expected synaptic hops to reach **TL1**, **EP**, and most **LNO** types, whereas ORN input reached most **FBT** and **PFL** types sooner.

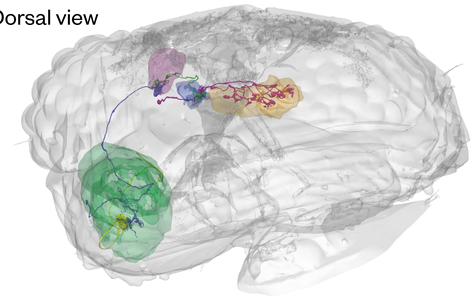


Information flow defines hierarchical layers

Layered pathway reconstruction identified the candidate intermediate neurons and routes that may support these modality-specific arrival patterns.



Dorsal view



R7p → MeTu → TuBu → TL1a

In the connectome reconstruction, this pathway follows the same anatomical organization identified by our analysis.

Together, these results reveal a layered hierarchy through which information reaches the CX, suggesting how *Megaphragma* may efficiently integrate sensory cues to guide behavior.

Conclusion

By combining these methods with the full connectome, we derive functional insights from the anatomy of a large, recurrent network without requiring detailed simulations.

References

[1] Honkanen A, et al. (2019) The insect central complex and the neural basis of navigational strategies. *J of Exp Bio* 222:jeb188854

[2] Hoeller J, et al. (2025) The organization of visual pathways in the Drosophila brain. *bioRxiv* 2025.12.22.696097.

[3] Schlegel P, et al. (2021) Information flow, cell types and stereotypy in a full olfactory connectome. *eLife* 10:e66018.

[4] Yin Y, et al. (2025) The Connectome Interpreter Toolkit. *bioRxiv* 2025.09.29.679410.

Scan QR code to explore all sensory pathway diagrams